Year 9 Mathematics

Unit 3: Sequences and Algebra

3.4 Factorising

Factorising

STARTER · 5 MIN

Date: _____ **Title: Factorising**

LEARNING OBJECTIVES

- Factorise $x^2 + bx + c$ into two brackets
- Factorise when there are negative numbers
- Factorise perfect squares and differences of two squares

STARTER *Answer in your book. No calculators.*

1. Write the factor pairs of 12
2. Two numbers multiply to 10 and add to 7
3. Expand $(x + 2)(x + 3)$
4. Work out -3×4

Starter — Answers

STARTER

Tick or correct in green pen

1. $1 \times 12, 2 \times 6, 3 \times 4$
2. 2 and 5
3. $x^2 + 5x + 6$
4. -12

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can factorise $x^2 + bx + c$ into two brackets	☐
SILVER	I can factorise when there are negative numbers	☐
GOLD	I can factorise perfect squares and differences of two squares	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Factorising

INTRODUCTION · 3 MIN

Factorising is expanding backwards.

The rule

Find two numbers that multiply to c and add to b.

$x^2 + 7x + 12 = (x + 3)(x + 4)$

Signs

+c: same signs. −c: different signs.

$x^2 + 2x - 15 = (x + 5)(x - 3)$

Perfect square

The two numbers are the same.

$x^2 - 10x + 25 = (x - 5)^2$

Difference of squares

No x term.

$x^2 - 49 = (x + 7)(x - 7)$

Examples 1 and 2 — Factorising $x^2 + bx + c$

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint two numbers: multiply to c, add to b

Example 1

Factorise

$x^2 + 7x + 12$

Example 2

Factorise

$x^2 + 9x + 18$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Multiply to 12, add to 7

$1 \times 12,\ \ 2 \times 6,\ \ 3 \times 4$

$3 + 4 = 7$

$(x + 3)(x + 4)$

Tip: expand to check your answer.

Example 2

Multiply to 18, add to 9

$1 \times 18,\ \ 2 \times 9,\ \ 3 \times 6$

$3 + 6 = 9$

$(x + 3)(x + 6)$

Tip: list the factor pairs in order.

Task 1

PRACTICE · 9 MIN

BRONZE *Factorise*

Hint two numbers: multiply to c, add to b

1. $x^2 + 5x + 6$

2. $x^2 + 7x + 10$

3. $x^2 + 8x + 15$

4. $x^2 + 9x + 20$

5. $x^2 + 11x + 24$

★ $x^2 + 13x + 36$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $(x + 2)(x + 3)$

2 $(x + 2)(x + 5)$

3 $(x + 3)(x + 5)$

4 $(x + 4)(x + 5)$

5 $(x + 3)(x + 8)$

★ $(x + 4)(x + 9)$

Examples 3 and 4 — Factorising with Negatives

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint +c: same signs −c: different signs

Example 3

Factorise

$x^2 - 8x + 15$

Example 4

Factorise

$x^2 + 2x - 15$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

+15: same signs, both −

$-3 \times -5 = 15$

$-3 + -5 = -8$

$(x - 3)(x - 5)$

Tip: the middle term is −, so both are −.

Example 4

−15: different signs

$5 \times -3 = -15$

$5 + -3 = 2$

$(x + 5)(x - 3)$

Tip: the bigger number takes the sign of b.

Task 2

PRACTICE · 9 MIN

SILVER *Factorise*

Hint +c: same signs −c: different signs

1. $x^2 - 7x + 12$
2. $x^2 - 9x + 20$
3. $x^2 + x - 12$
4. $x^2 - 2x - 8$
5. $x^2 - 3x - 28$

★ $x^2 + 5x - 36$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $(x - 3)(x - 4)$

2 $(x - 4)(x - 5)$

3 $(x + 4)(x - 3)$

4 $(x - 4)(x + 2)$

5 $(x - 7)(x + 4)$

★ $(x + 9)(x - 4)$

Examples 5 and 6 — Perfect Squares and Differences of Squares

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $(x + a)^2 = x^2 + 2ax + a^2$ $x^2 - a^2 = (x + a)(x - a)$

Example 5

(2 marks)

Is $x^2 - 10x + 25$ a perfect square? Give a reason.

Example 6

(1 mark)

Factorise

$x^2 - 49$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Multiply to 25, add to −10

$-5 \times -5 = 25, \quad -5 + -5 = -10$

$(x - 5)(x - 5)$

Yes: it is $(x - 5)^2$

Tip: both numbers are the same.

Example 6

$x^2 - a^2 = (x + a)(x - a)$

$49 = 7^2$

$(x + 7)(x - 7)$

Tip: there is no x term.

Task 3

PRACTICE · 9 MIN

GOLD

Questions 1–3: factorise, or say if it is a perfect square. 4–6: factorise

Hint

$(x + a)^2 = x^2 + 2ax + a^2$

$x^2 - a^2 = (x + a)(x - a)$

1 $x^2 + 6x + 9$

2 $x^2 - 14x + 49$

3 Is $x^2 + 8x + 12$ a perfect square?

4 $x^2 - 16$

5 $x^2 - 100$

★ $x^2 - 81$

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $(x + 3)^2$

2 $(x - 7)^2$

3 No: $(x + 2)(x + 6)$

4 $(x + 4)(x - 4)$

5 $(x + 10)(x - 10)$

★ $(x + 9)(x - 9)$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Factorise $x^2 + 6x + 8$

SILVER Factorise $x^2 - x - 6$

GOLD Factorise $x^2 - 36$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $(x + 2)(x + 4)$

SILVER $(x - 3)(x + 2)$

GOLD $(x + 6)(x - 6)$

Exam Practice — Factorising

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint two numbers: multiply to c, add to b

Question 1 (2 marks)

Factorise $x^2 + 7x + 6$

.....

Question 2 (2 marks)

Factorise $x^2 + 10x + 21$

.....

Exam Practice — Negatives

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint +c: same signs −c: different signs

Question 3

(2 marks)

Factorise $x^2 - 9x + 14$

.....

Question 4

(2 marks)

Factorise $x^2 + 3x - 18$

.....

Exam Practice — Special Cases

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint

$(x + a)^2 = x^2 + 2ax + a^2$

$x^2 - a^2 = (x + a)(x - a)$

Question 5

(2 marks)

Ali says, '$x^2 - 12x + 36$ is a perfect square.'

Is he correct? Give a reason.

.....

Question 6

(2 marks)

Factorise.

(a) $x^2 - 25$ (1)

.....

(b) $x^2 + 18x + 81$ (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(x + 1)(x + 6) B2

Tip: 1 × 6 = 6 and 1 + 6 = 7

Question 2

(x + 3)(x + 7) B2

Tip: 3 × 7 = 21 and 3 + 7 = 10

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(x − 2)(x − 7) B2

Tip: −2 × −7 = 14, −2 + −7 = −9

Question 4

(x + 6)(x − 3) B2

Tip: 6 × −3 = −18, 6 + −3 = 3

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$(x - 6)(x - 6)$ B1

Yes: it is $(x - 6)^2$ B1

Tip: $-6 \times -6 = 36$, $-6 + -6 = -12$

Question 6

(a) $(x + 5)(x - 5)$ B1

(b) $(x + 9)^2$ B1

Tip: (b) $9 \times 9 = 81$, $9 + 9 = 18$